

SECOND SEMESTER DIPLOMA EXAMINATION IN CHEMICAL ENGINEERING

FUNDAMENTALS OF CHEMICAL ENGINEERING

Maximum marks: 100

Time: 3 hrs

PART-A

(Maximum marks: 10)

- I
- a) Name and location of the two soap factories in Kerala?
 - b) State two advantages of size reduction
 - c) List any two heat exchanging apparatus
 - d) Define hydration
 - e) List any two temperature measurement equipment
- 5x2=10

PART-B

(Maximum marks: 30)

- II **Answer any five of the following question. Each question carries 6 marks**

- a) List the duties and responsibilities of a chemical engineer in an industry 6
- b)
 - i) In a double effect evaporator plant the second effect is maintained under vacuum of 475 torr(mm Hg).find the absolute pressure in kPa 3
 - ii) Iron metal weighing 500 lb occupies a volume of 29.25 L. calculate the density of iron in kg/L 3
- c) Discuss about the distillation process with an example 6
- d) State about the modes of heat transfer 6
- e) State the following with an example
 - i) Alkylation 3
 - ii) Esterification 3
- f) Illustrate the function of a manometer with a sketch 6
- g) List the scales of the following
 - i) Temperature 3
 - ii) pressure 3

5x6=30

PART-C

(Maximum marks: 60)

(Answer one full question from each unit. Each question carries 15 marks)

Unit-1

- III a) Do the following conversions
- | | | |
|-----|--|---|
| i) | 5 N H_3PO_4 to g/L | 3 |
| ii) | 3M H_2SO_4 to Normality | 3 |
| b) | List any six major oil refineries in India | 6 |
| c) | How many grams of carbon are present in 600gm of CaCO_3 | 3 |

Or

- IV a) List the product and raw material of any four chemical industries 8
- b) State the principle of the following for chemical industries:-
- | | | |
|-----|-------------|-----|
| i) | Economy | 3.5 |
| ii) | feasibility | 3.5 |

Unit-II

- V a) Compare absorption and desorption 8
- b) Explain extraction and leaching 7

Or

- VI a) Explain heat exchanging equipments 7
- b) List the methods of handling storage of liquids 8

Unit- III

- VII a) State the following:-
- | | | |
|------|---|---|
| i) | nitration | 2 |
| ii) | chlorination | 2 |
| iii) | Calcination | 2 |
| iv) | polymerization | 2 |
| b) | Explain cracking and pyrolysis with example | 7 |

Or

- VIII a) Define the following with example
- | | | |
|------|--|---|
| i) | conversion | 3 |
| ii) | yield | 3 |
| iii) | reaction efficiency | 3 |
| b) | Draw and describe about flow diagram based on industry | 6 |

Unit - IV

- | | | | |
|----|----|---|---|
| IX | a) | Describe flow measurement using Rotameter with a figure | 8 |
| | b) | Explain the measurement of pressure using Bourdon tube. | 7 |

Or

- | | | | |
|---|----|--|---|
| X | a) | Explain the requirement of Personal Protective equipments. | 8 |
| | b) | State the function of mechanical filter respirator | 7 |